Weathering the Worry

An Analysis of the Effect of Storm "Boris" on Climate Concern in Austria

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Submitted to Central European University - Private University
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In partial fulfilment of the requirements for the degree of
Masters in Economics, Data, and Policy

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Vienna
June 2026

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[1]

For bibliographic and reference purposes, this thesis should be referred to as: Burtscher, Katharina. [2026]. Weathering the Worry. An Analysis of the Effect of Storm "Boris" on Climate Concern in Austria. MA thesis, Department of Economics, Central European University - Private University, Vienna.

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Author's declaration

I, Katharina Burtscher, candidate for the MA degree in Economics, Data, and Policy declare herewith that the present thesis titled "Weathering the Worry. An Analysis of the Effect of Storm "Boris" on Climate Concern in Austria" is exclusively my own work, based on my research and only such external information as properly credited in notes and bibliography. I declare that no unidentified and illegitimate use was made of the work of others, and no part of the thesis infringes any person's or institution's copyright. I also declare that no part of the thesis has been submitted in this form to any other institution of higher education for an academic degree.

Vienna, June 8th 2026 Katharina Burtscher

Declaration of Generative AI

Generative artificial intelligence (GenAI) was used in this work. I, Katharina Burtscher, have reviewed and edited the output as needed and take full responsibility for the content, claims, and references. I used Gemini to

- Scrape newspaper articles about the September 2025 Central European flood and perform text processing with the "gemini-3-flash-preview" model.
- Assist my coding for the data processing and analysis, including data formatting tips and touch-ups for graphs.
- Support the writing process by providing grammatical corrections and stylistic improvements.
- Support in Latex formatting and troubleshooting.

Vienna, June 8th 2026 Katharina Burtscher

Abstract

With many climate mitigation policies lacking broad public support, understanding how extreme weather events influence climate concern is crucial for effective policy design and communication strategies. Consequently, my thesis investigates how the 2024 Central European floods affected climate worries in Austria, and whether the impact varies across regions with differing levels of exposure. To do so, I implement a difference-in-differences (DiD) design utilizing the 2023 and 2024 waves of the Austrian Socio-Economic Panel (ASEP), treating storm "Boris" as a natural experiment. Ultimately, the empirical results reveal that climate worry remained largely unaffected by the flooding, even in severly hit municipalities. This finding is consistent across different model specifications, robustly demonstrating that individuals do not alter their climate worry, despite living in a disaster-struck area.

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1. Introduction

Natural disasters have devastating impacts on residential infrastructure, local economies, and the environment. They are deeply disruptive, traumatizing events that destroy livelihoods and, in severe cases, cause fatalities. As climate change accelerates, these events are becoming increasingly more frequent and severe. Researchers estimate that if global temperatures rise to approximately 3°C above pre-industrial levels, the proportion of the global population exposed to catastrophic flooding and extreme heatwaves will multiply, potentially displacing hundreds of millions of people (Rahmstorf, 2024). In Austria, heavy 4-day rainfall events are now already twice as likely and 10% more severe, while a warming of 2°C is projected to double the number of uncomfortably hot days (Adil et al., 2025; Lizana et al., 2026).

Despite these prospects, only 11.5% of Austrians state that they would be willing to pay higher taxes to combat climate change, whereas a significant majority finds this measure unacceptable (ASEP, 2026). While increased taxes are rarely a popular proposition, the number of individuals that would be willing to reduce their standard of living in support of the environment is only slightly higher. At the same time, overall belief in climate change is high: only 1.6% of the Austrian population denies the existence of climate change (ASEP, 2026). Thus, there seems to be a discrepancy between climate change belief and the awareness about the urgency to take action against global warming. This raises the question of whether personally experiencing a climate-related natural disaster could bridge this gap. To study this link, I use storm "Boris", the Central European flooding event of September 2024, as a precedent. The following questions are central to the analysis:

1. *To what extent has the 2024 flooding event affected the worries about climate change in Austria?*
2. *Does the effect differ between more or less strongly affected areas?*

While climate worry is not a direct measure of policy support, it suggests that global warming is perceived as a serious threat (Arikan and Gunay, 2020). Studies show that individuals that exhibit more worry are also more likely to support climate policies (Harring and Sohlberg, 2017), even though it

is only one factor among others (Fransson and Gärling, 1999). In Austria, the relationship between climate concern and policy support also holds: Of those willing to pay higher taxes for environmental measures, 99.6% state that they are worried or very worried about climate change in 2024 (see figure 4.2 for reference).

Climate disasters are extreme weather events that become increasingly more frequent and intense due to climate change. Examples are floods, droughts, hurricanes and extreme temperatures. While the effect of climate disasters on climate worry has already been thoroughly researched (Hamilton et al., 2016; Zanocco et al., 2019; Ogunbode et al., 2020; Baccini and Leemann, 2021; Bulut and Samuel, 2024), the results differ, some studies concluding that experiencing climate disasters increases concern about the climate, while others find no such effect. This thesis adds to the existing literature by looking at the Austrian context, which has not yet been researched. I furthermore use the ASEP household survey, which is a new data source that provides the longitudinal panel data necessary for this analysis.

The regression results displayed in section 6.1 show that storm "Boris" had little to no effect on climate concern in Austria. Not only is the general trend in climate worry stagnant from 2023 to 2024, but the flood affected regions also show no change over the two years. Overall, this finding suggests that the firsthand experience of climate-related environmental disasters does not significantly increase climate worries in Austria.

This thesis is structured as follows: I start by reviewing the theoretical tie between climate worry and flood experience, as well as discussing previous literature in chapter 2. Chapter 3 provides an explanation of the flooding event, and chapter 4 offers an overview of the data. Next, chapter 5 describes the empirical strategy employed, highlighting the details and limitations behind the model specifications. Chapter 6 covers the results, as well as discussing the empirical shortcomings and robustness checks. Finally, chapter 7 provides general conclusions, policy implications and suggestions for further research.

2. Climate Disasters: Impact on Concern

This chapter gives an overview of the theoretical considerations and previous research on climate disaster experiences and their effect on worries and attitudes. Overall, existing literature finds mixed results, with most papers identifying only short-term increases in climate concern.

2.1 Linking Flood Experience to Climate Worry

Climate concern can be defined as the attitude towards environmental measures, information and behavior (Fransson and Gärling, 1999). Concern may either be driven by worry about the conservation of nature (ecocentric value orientation), the implicit threat to human well-being (anthropocentric altruism value orientation), or one's own safety (Fransson and Gärling, 1999). Worry and concern are two closely related concepts that will be used interchangably in this thesis.

Extreme weather events can increase climate worry through three channels: psychological proximity, social salience, and affective responses. First, because individuals often perceive climate change as a distant threat affecting other regions or future generations, experiencing a local extreme weather event reduces this psychological distance and renders the threat personally relevant (Sisco, 2021). Second, such disasters increase the salience of global warming in media coverage and private discourse, increasing public concern. Finally, these events can trigger affective responses; for instance, Bergquist et al. (2019) demonstrate that experiencing a disaster, such as a hurricane, induces negative emotional responses that drive heightened climate anxiety.

In particular, individuals who downplay climate change or deny its anthropogenic origins may be receptive to the experiential learning provided by a local flooding. Such extreme weather events have the potential to convey the existential threat of climate change to skeptics more convincingly than scientific facts. However, the research on this is divided - while Sambrook et al. (2021) point out that prior beliefs continue to heavily influence how people interpret flooding in Germany, Chen et al.

(2024) analyze responses to hurricane "Harvey" and the 2021 Texas winter storm, finding that Republicans updated their climate attitudes and narrowed the ideological gap to Democrats. In the Austrian context, climate attitudes also follow a classic left-right relationship: FPÖ (far-right) supporters are significantly less worried about climate change and less likely to support environmental policies than voters of other political parties (see appendix G). Thus, the 2024 floods provide an natural experiment to test whether localized, firsthand disaster experience can override prior beliefs within the Austrian population.

2.2 Effects of Climate Disasters on Public Attitudes: A review of the Literature

This section provides a synthesis of the relevant literature on the impact of climate disasters on climate worry. I will give an overview of the general findings, as well as highlighting different thematical foci (impact on voting behavior, geographic proximity, temporal proximity). Furthermore, I will discuss the methodological approach employed by Bulut and Samuel (2024), as it relates to the empirical design used in this thesis.

In their study of climate awareness change in response to living in a disaster-struck area in the US, Konisky et al. (2016) identify positive, albeit moderate, effects. On the other hand, Zanocco et al. (2018) find that respondents in more affected regions claim to have experienced more harm, but do not consistently change in their level of climate concern. Similarly, Hamilton et al. (2016) also observe no increased climate worry in response to flooding in the US. Consequently, the field remains divided, with some studies finding positive and others negligible effects.

The impact of extreme weather events also extends to institutional and political behavior. Baccini and Leemann (2021) analyze the Swiss context, finding that municipalities hit by a natural disaster within 12 months of a climate-related referendum saw a 1 percentage point increase in support for climate measures, alongside with higher voter turnout. The unique direct-democracy structure of Switzerland allows for the isolation of climate-specific policy support. However, translating these findings to

other countries remains complex. In Austria, political affiliation is influenced by a wide array of confounding factors that would limit the interpretability of such an analysis. This provides motivation to focus on climate concern as the outcome variable rather than party preference.

Most studies find that the relationship between climate disasters and climate concern is only temporary. Konisky et al. (2016) conclude that the positive effects already fade one month after the event. Baccini and Leemann (2021) find that the effect of extreme weather on referendum voting behaviour diminishes after 10 months. Consequently, the survey period determines whether short-term or long-term effects are measured, with the latter most likely yielding no significant results.

Furthermore, the personal proximity to the disaster events is a relevant factor. Ogunbode et al. (2020) argue that the inconsistent findings in the exisiting literature partly stem from a failure to differentiate between individual affectedness (e.g., personal property damage) and local affectedness (e.g., damage to public infrastructure). Their research indicates that while personal experience significantly alters attitudes, local affectedness does not. In a follow-up paper analyzing their null-effect from 2018, Zanocco et al. (2019) come to a similar conclusion, finding no relationship between living in an affected county and supporting climate-mitigating policies, but a positive effect for those directly exposed to extreme weather events.

In their analysis of the 2021 floods in Luxembourg, Bulut and Samuel (2024) employ a similar approach to the one used in this thesis. They exploit a flooding event as a natural experiment, analyzing the effects on individuals' attitudes and behaviour depending on geographic proximity with a difference-in-differences set-up. The authors find that the relationship between pro-environmental attitudes and climate-friendly behavior was significantly stronger in affected areas. However, there are a few notable differences to my own work: For one, their study uses an age-restricted sample, which only considers individuals aged 16 to 29. Consequently, Bulut and Samuel (2024) findings are not generalizable to the broader population. As the authors themselves state, younger cohorts' attitudes are still more adaptive compared to older demographics. Furthermore, the study offers limited methodological transparency regarding the specific criteria used to define geographic "proximity" to

flooding, only stating that they divide the areas into binary categories of affectedness. However, this classification is not trivial; my methodology behind the categorization of affected areas is discussed extensively in sections 4.1, 4.2 and 5.2.

In short, research finds that 1) climate concern tends to rise following extreme weather events, yet this finding is not unanimous, 2) climate worry increases when individuals are personally affected by climate disasters, 3) increased concern is usually short-lived, and 4) in general, elevated worry leads to greater environmental policy support. This thesis considers short-term impacts of flooding on climate worry and aims to identify individuals in severely hit municipalities. My analysis relies on the Austrian Socio-Economic Panel (ASEP), which is a panel survey that is representative for the Austrian population. This enables me to draw conclusions about the impact of a climate disaster on climate concern in Austrian as a whole. Furthermore, the municipal-level information on climate affectedness allows me to identify flood exposure on a fine level.

3. Description of the Event

The following chapter gives an overview of the unfolding of storm "Boris", as well as its severity and immediate impacts. I argue that the flooding caused an unprecedented climate disaster for the most severely hit regions in Austria, as it exceeded all previous precipitation and water level records.

3.1 Meteorological Assessment

After a hot and dry summer, the Mediterranean low-pressure-cyclone “Boris” (in Germany: “Anett”) moved from Italy to Central and Eastern Europe, where it eventually dissipated from September 12 to 16, putting a sudden stop to the summer conditions (Greilinger et al., 2024). Nearly all of Austria experienced heavy rains over the course of five days between September 12 to 16. Starting in Southern Austria, “Boris” moved north and east on September 13th, causing record-level rainfall in Lower Austria, as precipitation was constant and heavy in this area. Overall, most of Austria experienced rainfall of approximately 100 mm per m^2, which is significantly higher than historical September averages of approximately 60-90 mm in many areas (GeoSphere Austria, 2026).

This is strongly contrasted by the situation in the areas that were at the epicenter of the rainfall, where flood levels as high as 400 mm were documented. “Boris” caused the heaviest five-day rain event ever recorded for the states of Lower Austria, Upper Austria, and Vienna, with an average of 25% more precipitation than in the previous most extreme flood from 1961 (Greilinger et al., 2024). The regions surrounding the districts of “Tulln” and “St. Pölten” recorded approximately 200 mm more precipitation over a five-day period than its previous record. In fact, until storm “Boris”, 5-day totals exceeding 275 mm had rarely been recorded outside the Alpine region, and totals exceeding 300 mm had never been documented before (Greilinger et al., 2024). While in large areas the flooding could be classified as a once-in-a-century event, said regions clearly passed the once-in-a-millenium benchmark.

3.2 Damage

Storm "Boris" has threatened the safety of the residents in the affected regions, as well as causing infrastructural and economic damage. As the strongest hit area, Lower Austria declared its entire territory a disaster zone. 20 of its dams were severely damaged (of which 9 fully breached), 21 000 buildings and 1000 firms were directly affected, 20 000 households were out of power and the drinking water system was interrupted in multiple municipalities. Tragically, more than 3000 people had to be evacuated, 24 experienced injuries, and 5 people died (Higer, 2025).

In a first assessment, Zibuschka (2025) estimated the overall cost of the flooding at €1.07 billion, more than two thirds of which was damage to private property. Beyond physical and economic loss, research shows that experiencing a flood can lead to post-traumatic stress disorder (PTSD), depression and increased anxiety (Mason et al., 2010). As a clerk at the Lower Austrian Department of Fire Brigades and Civil Defense and local fire fighter, Mr. Obermaißer, recounts, many people panicked when in 2025 water levels briefly rose during renewed heavy rainfall [1].

[1] Stefan Obermaißer (Employee, Office of the Lower Austrian Government. Department for Fire Brigades and Civil Defense), in discussion with the author, April 2026

4. Data

This thesis uses data from the Austrian Socio-Economic Panel (ASEP), municipal-level information on flood affectedness from the Lower Austrian Department for Fire Brigades and Civil Defense, as well as a municipality classification dataset, created on the basis of newspaper articles covering the event. The following section describes these three sources in more detail.

4.1 ASEP

The ASEP is a household panel that started as a pilot study in 2023. The survey is representative for the Austrian population and questions the same households every year. It is implemented by Statistics Austria; wave 1 (2023) and 2 (2024) are available for free download from AUSSDA (ASEP, 2024, 2026). For this thesis, the ASEP provides information on socio-demographic characteristics, as well as beliefs, attitudes and the outcome variable: worry about the consequences of climate change. Wave 1 (2023) has a sample size of N = 2978; wave 2's (2024) sample size is N = 1845. In total, 1392 people provided a valid response to the question about their climate concerns in both years. In 2024, the vast majority of respondents first opened the questionnaire environment in October, only a few weeks after storm "Boris. Consequently, the climate worry surveyed in in the second ASEP wave provides a measure of recent shifts in attitudes in response to the flood.

4.1.1 Outcome Variable

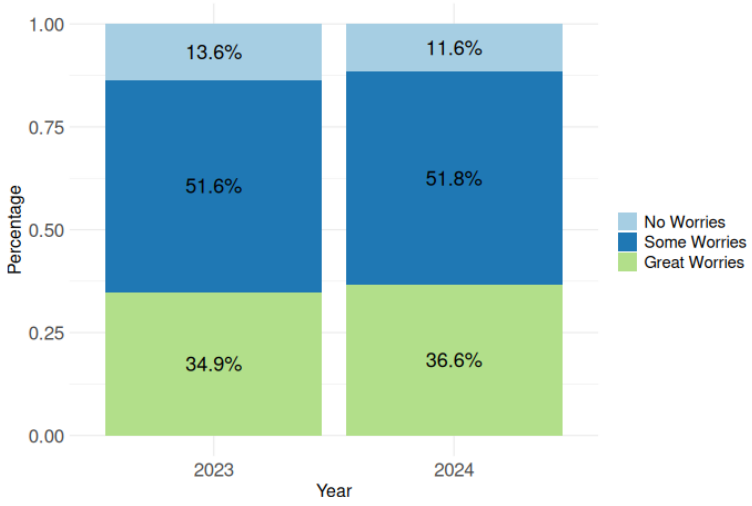
In the ASEP, climate worry is measured with the following question:

What about the following issues, are you worried about them? About the consequences of climate change.
No Worries / Some Worries / Great Worries

Overall, the majority of people indicated that they have at least "Some Worries" about climate change, with only 9% claiming that they were not worried in both waves. On the other hand, approximately one third stated that they have "Great Worries", suggesting that the variable faces a ceiling effect. The

implications of this are discussed in section 6.2.2.

Figure 4.1: Worry about Climate Change



Source: (ASEP, 2024, 2026)

4.1.2 Demographics and Climate Concern: A Descriptive Overview

Women are considerably more worried about the consequences of climate change than men: 36% of women say that they have "Great Worries", whereas only 26% of men express great concern. Men are also amost twice as likely to have no worries. Furthermore, there are significant differences between the Austrian federal states. Overall, Styria and Vienna are among the states that exhibit comparatively high worry. As the two states host the two largest cities, this relationship might also be driven by the rural-urban divide. Table 4.1 below shows climate worry pertaining to the degree of urbanisation (urban = cities; suburban = small cities and urban centres; rural = no urban centre). There is a strong correlation between level of urbanization and climate concern, with individuals living in the countryside being more than twice as likely to have no worries.

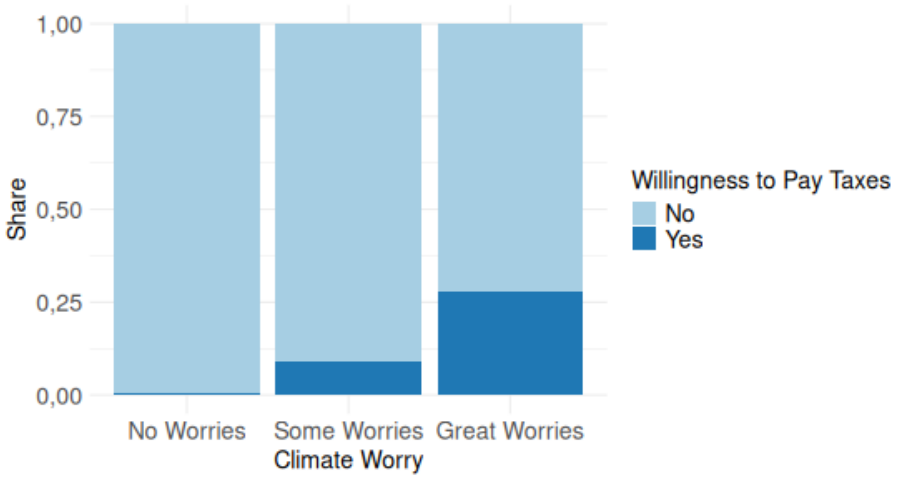
Table 4.1: Climate Worry by Degree of Urbanization

Urban Level	No Worries	Some Worries	Great Worries
urban	8.7%	56.6%	34.7%
suburban	14.4%	55.2%	30.4%
rural	19.5%	52.9%	27.5%

Source: (ASEP, 2026)

Lastly, there is a significant positive correlation between the willingness to accept higher taxes in support of environmental protective measures and climate worry. The graph below shows that almost none of those with no climate worries would be willing to pay higher taxes, whereas the share rises to over 25% among those that are very worried. This highlights the strong relationship between climate worry and policy support in Austria.

Figure 4.2: Willingness to Pay Higher Taxes for Environment by Climate Worry



Source: (ASEP, 2026)

4.2 Construction of the Treatment Group

While large areas in Austria were affected by storm "Boris", severe damage and flooding was still regionally distinct and even differed within districts and towns. The ASEP dataset allows for identification of the respondent's municipality of residence by the "Gemeindekennziffer" ("municipal code"), but as it doesn't survey the flood affectedness directly, this information has to be sourced elsewhere.

4.2.1 Property Damage Indicator

The Lower Austrian government has collected self-reported data from the municipalities on the number of flooded residential, municipal and commercial buildings (NÖ Landesregierung, 2024). For this thesis, only the damages to residential property are taken into consideration. The reason for this is twofold: First, residential property damage is a phenomenon most likely experienced firsthand by survey respondents. Second, it strongly correlates with harm to both commercial and municipal infrastructure; therefore, including additional damage data would duplicate information. From the raw count of damaged buildings, I construct a *damage_indicator*. For this, the level of affectedness is calculated simply by dividing the number of damaged residential buildings per municipality by its population (retrieved from STATcube (2026) for the 2024 population counts), and multiplied by 10 for readability.

$$\text{damage_indicator}_m = \frac{\text{Damage}_m}{\text{Population}_m} * 10 \tag{4.1}$$

In total, 151 individuals live in a municipality where at least one residential building was harmed. The *damage_indicator* is a straightforward measure of the severity of the impacts of storm "Boris" for each municipality. It reflects the “average” flood experience of the local population, since flooded basements and other minor property damage in private homes are common, as opposed to non-residential harm, such as crop failures or dam breaches. However, because disaster relief and monitoring is administered on the state level, this data is only available for Lower Austria. While it was hit hardest, this categorization inevitably misses affected regions outside of Lower Austria.

4.2.2 Media Index

To generate a measure that covers all of Austria, I additionally created a media index, which I constructed as a weighted count of all municipalities that appear in newspaper articles that contain information on storm "Boris". Using the keywords "Hochwasser" (German: flood) and "2024", I scraped 800 articles from Heute (Heute, 2026), 281 articles from ORF (ORF, 2026) and 100 articles from MeinBezirk (Bezirk, 2026) all of which are news portals that cover local events. I further limited the articles to a timeframe between September 1st 2024 to December 31st 2024, which left 588 articles re-

maining. I then analyzed the bodies of the articles with a large language model. The AI was instructed to only extract towns and districts that were directly affected by the flood and count their number of occurrences. The applied prompt can be found in appendix B.

Using the raw number of mentions would disproportionately overrepresent densely populated areas, which dominate media reports. To control for population differences while mitigating the arbitrary nature of low frequencies, where a town mentioned twice is not necessarily twice as affected as a town mentioned once, I apply a logarithmic transformation. Specifically, the *media_index* is calculated by dividing the logarithm of a municipality's flood mentions by the logarithm of its population. This approach controls for population differences while smoothing out the volatility of low counts.

$$\text{media_index}_m = \frac{\log(\text{flood_count}_m + 1)}{\log(\text{population}_m)} \tag{4.2}$$

Both the *media_index* and the *damage_indicator* allow for continous treatment.

5. Empirical Strategy

To analyze if storm "Boris" affected climate worries in Austria, I employ a difference-in-differences (DiD) approach. Because of its panel structure, the ASEP surveys climate worry in both 2023 and 2024 for the same individuals, allowing for a before-and-after comparison. Within the sample there is regional variation, which I exploit for the comparison of people in strongly affected (treated) communities with those in largely unaffected (control) regions. DiD is an appropriate identification method in this set-up because storm "Boris" was an exogenous event; the fact that it hit some regions stronger than others was random and could not have been foreseen more than a few weeks out. Furthermore, the control and treatment regions are different from each other: most of the affected areas were less densely populated than the Austrian average and had lower climate worry to begin with. As DiD can handle both time-invariant differences between groups and time-variant changes across the entire sample, it is a suitable approach to analyze the effects of storm "Boris" on climate worries.

DiD follows the main assumption that control and treatment group would have followed the same trend in the absence of treatment. Thus, had storm "Boris" not occurred, I assume that affected and unaffected regions would have remained different in their levels of worry, but their trajectory would have aligned. Because I only observe climate worry in two datapoints (2023 & 2024), it is not possible to test the parallel trends with the same data that underlies the analysis. Instead, I use *"Worry about Climate Change"* from the European Social Survey (ESS) which is available for Austria in 2016, 2020 and 2023 (see appendix C for more details) (European Social Survey, 2016, 2020, 2023). Plotting climate worry of Lower Austria against the rest of country over time shows that the two groups closely follow each other. Conducting a parallel trends test also yields insignificant results. Unfortunately, the ESS dataset does not allow for a detailed differentiation between regional units, as it only includes state-level information on the respondents' residence. Consequently, the parallel trends test can only be validated for model (1) (see section 5.2 and table 6.1). The issues this might cause are discussed in section 6.2.1.

5.1 Main Specification

Ordinary Least Squares (OLS)

The DiD approach is implemented with two different estimation methods. First, I calculate an ordinary least squares (OLS) model, with the following specification:

$$\text{wor_clima}_{it} = \alpha_i + \beta_1 \text{Post}_t + \beta_2 \text{Treated}_i + \beta_3 (\text{Post}_t \times \text{Treated}_i) + \mathbf{X}_{it} + \epsilon_{it} \tag{5.1}$$

where i denotes indivduals and t denotes time. *Post* is a dummy that captures the change from 2023 to 2024, *Treated* is the binary or continuous treatment value, and X is a vector of relevant controls. Furthermore, β_3 captures the DiD coefficient. OLS models assume homoskedasticity and that the error terms are independent and identically distributed (i.i.d.). This assumption is relaxed through the use of cluster-robust standard errors on the individual level, which allows arbitrary correlation within each respondent. Furthermore, OLS treats the three-level ordinal variable as a continuous measure. This implicitly assumes that the intervals between categories are equidistant, which means that the shift from "No Worries" to "Some Worries" is the same as the shift from "Some Worries" to "Great Worries". Therefore, as an additional estimation method, I calculate an Ordered Logistic Regression. As the results in section 6.1 and appendix D show, the OLS and logit regression estimates are largely consistent.

Ordered Logistic Regression

To account for these issues, I further estimate an ordered probability model:

$$\ln\left(\frac{P(\text{wor_clima}_{it} \leq k)}{P(\text{wor_clima}_{it} > k)}\right) = \alpha_k + \beta_1 \text{Post}_t + \beta_2 \text{Treated}_i + \beta_3 (\text{Post} \times \text{Treated})_{it} + \mathbf{X}_{it}\Gamma \tag{5.2}$$

where the left-hand side of the equation represents the log-odds of a respondent falling into or below a given climate worry category k, versus the cumulative probability of reporting a higher level of worry. α_k respresents the category-specific cutpoints and X_i are relevant controls. On par with the OLS model, the ordered logit is estimated with clustered standard errors on the individual level.

Ordered logit models rely on the proportional odds assumption, which states that the relationship between the independent variables and the logits is the same for all cutpoints. Testing this using a Brant test yields insignificant results ($p = 0.54$), which suggests that the proportional odds assumption is not violated.

Controls

Socio-demographic characteristics are known to be among the most important predictors for climate attitudes (Harring and Sohlberg, 2017; Peñasco and Grossman, 2026; Poortinga et al., 2019; Weckroth and Ala-Mantila, 2022). Based on theoretical findings, the OLS and ordered logit models include education, age, gender, level of urbanization and party preference as controls. For the purpose of identifying relevant controls, I further trained an XGBoost model, which additionally identified "anxious personality" as an explanatory parameter. As a matter of fact, it strongly correlates with the outcome and it was consequently added to the models. Income showed no substantial correlation with climate worry and it was therefore not included. Results from sensitivity checks are displayed in appendix E.

5.2 Models

The following section describes the various implementations of the OLS and ordered logit model. While the overall model structure was kept constant throughout, the treatment group changes for each model. Each treatment definition has its advantages and drawbacks and compensates for the specific limitations of the others.

Model 1: Binary Treatment

The first stage of the analysis involves simply defining all residents of Lower Austria as the treatment group. It was by far the most severly affected state in Austria and its entire territory was declared as a disaster zone. While not all households in Lower Austria were directly affected by the flood, those that weren't still received the same governmental information, as disaster relief, communication and financial reimbursements were organized on the state level. This motivates choosing all Lower Austrian residents to be treated.

While the binary treatment is a crude measure of affectedness, it has certain benefits. For one, it is straightforward and easy to implement. Furthermore, this specification allows for the assumption of parallel trends between treatment and control. On the downside, choosing Lower Austria as treated results in a blurry treatment group. Not all individuals in Lower Austria were affected by storm "Boris", while some regions outside of the state also experienced flooding. The limitations this causes are discussed in section 6.2.1.

Model 2: Continuous Treatment (Damage) - Lower Austria only

The second model uses the *damage_indicator* (as defined in formula 4.1), which measures the number of residential buildings damaged in a municipality, weighted by its population. Because this data is only available for Lower Austria, the sample in Model (2) is limited to Lower Austrian residents. Each respondent is assigned a *damage_indicator*, based on their municipality of residence. If a person moved between 2023 and 2024, they were given the indicator value for the later address. The *damage_indicator* thus serves as continuous treatment, where respondents that live in a town with a larger number of damaged buildings are assigned a higher treatment value. It is a more nuanced measure of treatment than Model (1), but has two major drawbacks: For one, restricting the sample to Lower Austrians tremendously reduces the sample to N = 251. Consequently, estimations are unlikely to yield significant results. Secondly, the parallel trends assumption cannot be validated for this model.

Model 3: Continuous Treatment (Damage)

Model (3) employs the same treatment group as Model (2), with the difference that the rest of the country is used as control in addition to those in Lower Austria that are unaffected by the flood and have a *damage_indicator* of 0. This improves Model (2) because it increases the sample size, while maintaining a refined measure of flood affectedness. However, the treatment group excludes individuals that lived in flooded municipalities outside of Lower Austria, which leads to an attenuation bias, as discussed in section 6.2.1.

Model 4: Continuous Treatment (Media Index)

Lastly, Model (4) uses the value of the *media_index* (calculated in formula 4.2) as continuous treatment. The model exploits a fine-grained measure of flood affectedness, while as the only model, also considering hit areas outside of Lower Austria. The drawback of the model is its reliance on media coverage, which is biased to capture more spectacular happenings (such as bridges collapsing or dam breaches) over more "standard" flooded basements and fields. Furthermore, any town not mentioned in the articles will not be captured, regardless of the actual flood impact. Thus, the *media_index* can only serve as a proxy of flood affectedness.

6. Results

This chapter discusses the regression results and its implications. Results are shown for all model specifications described in section 5.2.

6.1 Empirical Findings

Table 6.1 displays the OLS regression results for all four model specifications. Each model was calculated with a full set of controls and cluster-robust standard errors on the level of the individual. The table shows the estimates of the year-effect (*Post-Treatment*), the level difference between treatment and control group (*Treated Group*), the DiD estimator (*DID (Post x Treated)*) and the number of observations. The actual sample size is half of the displayed *Num.Obs* due to the long format of the underlying dataframe. Model (1) refers to the model with Lower Austria as binary treatment, model (2) uses the continuous *damage_indicator* as treatment, while restricting the sample to Lower Austrians. Model (3) builds on model (2), adding the rest of Austria to the control group and model (4) applies the *media_index* as continuous treatment. For more detailed description of the models please consult section 5.2. The table below displays the OLS results; the summary output of the ordered logit model can be found in appendix D.

Table 6.1: OLS Summary Statistics

	Model 1	Model 2	Model 3	Model 4
Post-Treatment	0.018	-0.065	0.004	0.007
	(0.020)	(0.042)	(0.018)	(0.019)
Treated Group	-0.020	-0.118	-0.136	-0.051
	(0.044)	(0.168)	(0.147)	(0.293)
DID (Post x Treated)	-0.077+	0.024	-0.065	-0.172
	(0.042)	(0.185)	(0.168)	(0.331)
Num.Obs.	2620	502	2620	2620

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Note: Standard Errors displayed. Controls: age, gender, education, party preference, urban level, anxious character. Cluster-Robust Standard Errors.

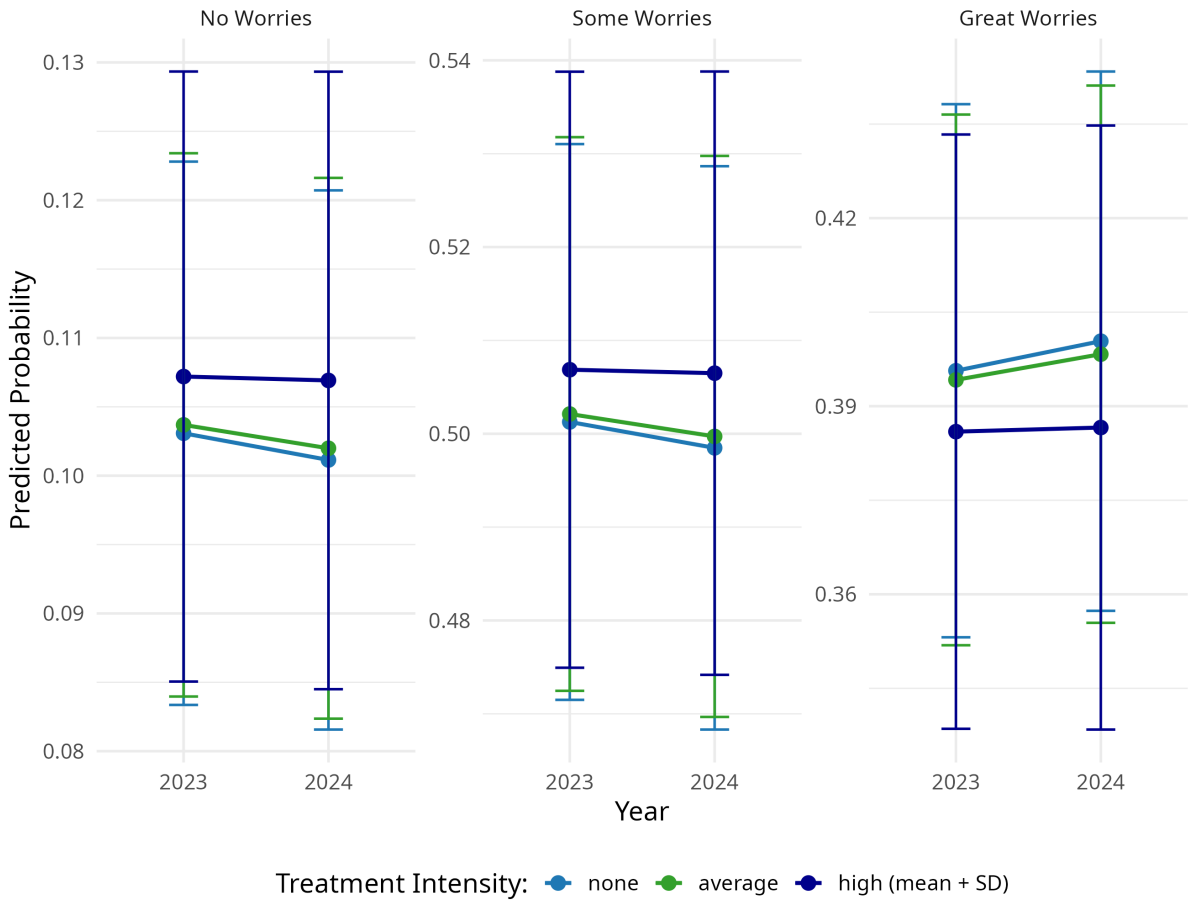
Source: (ASEP, 2024, 2026)

The OLS results clearly show that storm "Boris" did not have significant impact on climate worries. A comparison of the four models further reveals that the DiD estimates of model (1), (3) and (4) imply that being treated in fact leads to a minor decline in climate worry. However, the DiD estimate is consistently close to 0 and not significant at the 5% level. It is important to keep in mind that model (2) uses a different underlying sample size (restricting the sample to Lower Austrians), which might partly cause insignificant results. Furthermore, each of the models uses a different definition of treatment and/or control. While all of the models have important limitations, the fact that they unanimously show null-results supports the claim that the flooding event had no effect on climate worries.

Figure 6.1 shows the probability of falling into each outcome level ("No Worries", "Some Worries" or "Great Worries"), using the ordered logit estimation results of model (3) (see appendix D.1). Each "column" represents the probability for one of the three outcome levels. Because the treatment variable in specification (3) is continuous (*damage_indicator*), I evaluate its effects by constructing three scenarios: Group 1 represents the baseline probability of each outcome in the absence of treatment (no damage); group 2 estimates these probabilities for an individual experiencing an average level of treatment (mean damage). Finally, Group 3 models a high-treatment scenario, defined as a damage level one standard deviation above the sample mean.

Figure 6.1 displays the change in the response behavior from one 2023 to 2024, differentiated by the three groups. Please note that the scales on the y-axis have been magnified to highlight small differences.

Figure 6.1: Model 3 - Treatment Effect on Climate Worry



Source: (ASEP, 2024, 2026)

While the probability for choosing "No Worries" or "Some Worries" is larger among those with higher treatment intensity, it is lower for those who have "Great Worries". Thus, individuals that lived in an area that experienced above-average flood damages exhibit a lower level of worry both before and after the event. This aligns with the model (3) *Treated Group* estimate from table 6.1. Most importantly, the slopes of the three are largely parallel, which means that despite living in a region with more property damage, individuals' climate concern did not adjust accordingly.

6.2 Empirical Shortcomings

In this thesis, the treatment group identification and the ceiling effect of the outcome variable are two sources for bias in the estimates. Both issues are discussed in the section below.

6.2.1 Misclassification of Treatment

The identification of the treatment group is a challenge in this thesis, as there is no direct measure of flood affectedness in the ASEP dataset and the treatment group has to be defined based on external information. Therefore, models (2) and (3) use the *damage_indicator*, and Model (4) the *media_index* as an approximation for flood affectedness.

In model (4), the construction of the *media_index* introduces a non-trivial measurement error. First, the distribution of municipal mentions in newspaper articles is strongly right-skewed, with most towns appearing infrequently. For municipalities that are only mentioned a few times, marginal variations (e.g. one versus two mentions) artificially double the intensity metric. This issue is largely accounted for by logarithmizing the values in equation 4.2. Second, because the index relies strictly on media coverage, it functions as a proxy rather than a direct measure of flood affectedness. Consequently, this induces a misclassification of treatment assignment for a subset of observations within the dataset. Furthermore, models (1) and (3) limit treatment to Lower Austrians, effectively excluding individuals from flooded towns elsewhere. In all of my setups I argue that the misclassification is an error of exlusion (false negatives), rather than inclusion (false positives). This means that for models (1) and (3), individuals outside of Lower Austria are assumed not treated, even if they lived in a flooded region. For model (4) I expect that the towns mentioned in the articles were actually affected (see appendix B for the AI prompt used to select relevant municipalities), but not all affected municipalities were captured by a newspaper entry. In the case of a binary treatment variable, the one-sided misclassification takes the following form:

$$T = T^* * C \tag{6.1}$$

where C is a dummy that equals 1 if treatment is correctly classified, T^* is the true treatment group and T is the treatment group as specified in the model. This estimation only shows false negatives, which means that some treated individuals are misclassified as control, but no control individuals are misclassified as treated.

Negi and Negi (2025) analyze the econometric consequences of treatment misclassification in DiD designs, demonstrating that strictly false-negative, exogenous misclassification induces attenuation bias. In the context of this study, the assumption of exogenous misclassification is highly plausible, as residence in vs. just outside of Lower Austria can define flood-exposure classification, but does not correlate with climate worry. Methodologically, the attenuation bias emerges because the control group includes treated individuals, increasing the similarity between the two groups and biasing the estimated treatment effect toward zero (Negi and Negi, 2025). Luckily, estimates affected by an attenuation bias can still be interpreted. Finally, while Negi and Negi (2025) formalize this misclassification framework within a linear (OLS) context, the underlying logic of attenuation via sample contamination applies equally to the non-linear ordered logit specification. Consequently, any identified treatment effects can be interpreted as conservative lower bounds of the true impact.

6.2.2 Ceiling Effect

A ceiling effect occurs when the measurement of the outcome variable is not able to differentiate between individuals at the upper end of the distribution. This can lead to an understimation of the treatment effect, as individuals that are already at the top of the scale cannot become more worried, even if affected by the flood (Lozano et al., 2008). As climate worry is measured on a 3-point scale and one third of all responses in 2023 were already at the top of the range ("Great Worries"), this is particularly relevant in my case.

Overall, I have to assume that the reliability (degree to which the measurement is consistent and free from random error) and validity (degree to which the measurement accurately captures what it is intended to measure) of the outcome variable are impaired by the ceiling effect and that I have to interpret my results with caution. According to Howe (2018), previous beliefs about climate change

strongly impact the perception of extreme weather events. It is thus possible that individuals that were already very worried in 2023 are the ones that responded strongest to the flood, now being more worried than ever. However, because of the ceiling effect, this change is not captured by the point estimates. To assess the impact on my estimate, I conducted a robustness check, which is discussed in section 6.3 limits the sample to those with "No Worries" or "Some Worries" in 2023.

6.3 Robustness Checks

The first robustness check, as discussed in the section above, excludes indiviuals who already had "Great Worries" in 2023 from the sample. Hence, the resulting estimates should not be biased by the ceiling effect of the outcome variable. Second, I ran a Linear Probability Model (LPM), where those who have "No Worries" or "Some Worries" are assigned 0 and people with "Great Worries" are assigned 1. By binarizing the variable (0, 0, 1), the LPM explicitly measures the determinants of moving to the highest threshold, as well as enabling a more straightforward interpretation of the estimates. Finally, Model (3) shows the results for an alternate outcome variable - worry about the environment. The rationale being that if the respondents were impacted in their envionmental concern, but did not attribute the flood's extent to climate change, they might still exhibit increase in overall environmental worry. All three robustness checks use the *damage_indicator* as continuous treatement variable on the full Austrian sample (thus analogous to Model (3), as outlined in section 5.2).

Table 6.2: Robustness Checks

	Model 1	**Model 2**	**Model 3**
Post-Treatment	0.192***	-0.001	-0.015
	(0.021)	(0.013)	(0.018)
Treated Group	0.061	-0.131	0.114
	(0.102)	(0.079)	(0.164)
DID (Post x Treated)	-0.138	0.007	-0.192
	(0.156)	(0.095)	(0.173)
Num.Obs.	1664	2620	2614

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Note: Standard Errors displayed. Controls: age, gender, education, party preference, urban level, anxious character. Cluster-Robust Standard Errors.

Source: (ASEP, 2024, 2026)

Once again, the models show that storm "Boris" had no effect on climate worries. By omitting the top category in model (1), I removed the artificial ceiling. This unmasks that there is, in fact, a true upward basline trend overtime, which is why *Post-Treatment* becomes positive and statistically significant. If the ceiling effect had suppressed the treatment group disproportionately, the DiD coefficient would have changed as well. However, because the DiD estimate is stable, it reassures that the primary estimate is valid. Model (2) specifically measures the transition from low and medium concern to high concern, with all estimates being close to 0 and insignificant. Interestingly, the DiD estimate of model (3) is also insignificant, with a negative point estimate. This suggests that the primary null findings do not stem from a failure of the respondents to attribute storm "Boris" to climate change, but rather from the event's overall lack of impact.

7. Conclusion

The analysis finds no evidence that storm "Boris" increased climate concern among surveyed Austrians. Across sepcifications, the DiD estimator is statistically insignificant, point estimates are small and several are even negative, supporting a null effect. This result holds when varying the treatment definition, sample composition, and regression form; suggesting that living in a flooded region during the event did not meaningfully shift climate worry.

Several limitations constrain interpretation. The treatment measure captures residence in an affected region rather than individual-level exposure; only a minority of respondents likely experienced personal damage, given the right-skewed distributions of the *damage_indicator* and *media_index*. Prior research indicates that direct personal affectedness is a stronger predictor of climate concern than regional residence (Zanocco et al., 2019), so estimates here may understate effects on those actually harmed. The DiD strategy also rests on the parallel-trends assumption that may not hold: if pre-treatment trajectories differed systematically between affected and control areas (e.g. if rural regions had experienced sharper declines in worry in the absence of the flood), the interaction term is biased.

Several plausible mechanisms could account for the null findings. First, a "finite pool of worry" may have constrained attention: individuals coping with immediate, practical consequences of the flood (e.g. property cleanup, disrupted transport, etc.) may have prioritized short-term recovery concerns over more complex environmental risk. When investigating anxiety in Japan after the Tohoku earthquake and the Fukushima nuclear accident in 2011, Nakayachi et al. (2015) find that worry about earthquakes and nuclear hazards increased, while other worries almost universally decreased, supporting the finite pool of worry hypothesis. In the context of this thesis, "Boris" may have increased worry about future floods, while at the same time crowding out concern about climate change, which is a more abstract and long-term risk. Similarly, other competing societal stressors (e.g. Covid-19, geopolitical tensions) may have overshadowed climate worry. Furthermore, because the climate-worry item appears in a battery alongside other worries, respondents' relative ranking of concerns may

have shifted, even if absolute worry about climate change did not. Secondly, as the flood was only a momentary shock, it may not have been long-lasting enough to alter underlying beliefs. Moreover, despite the localized devastation it inflicted, storm "Boris" might not have crossed a severity level required to fundamentally shift public attitudes toward climate change. Lastly, affective responses to extreme weather events are not strictly rational or uniform: some individuals panic during renewed river rises while others subsequently construct residential buildings in flood-prone areas, illustrating heterogeneous behavioral and emotional reactions. [1]

The policy implications are consequential. If extreme weather exposure alone does not reliably increase public climate concern, then policymakers and advocates cannot rely on episodic disasters to generate broad shifts in public support for climate measures. Instead, sustained engagement strategies that connect climate change to individuals' lived experiences may be necessary to build durable concern and support for action. Efforts should therefore combine targeted outreach to affected communities (where personal experiences may be more salient) with broader, sustained communication strategies that link concrete climate policies to people's everyday priorities (health, safety, local infrastructure, economic security, etc.). Thus, messaging should aim to translate the experienced events into a narrative that connects the relationship between climate change and its impact on people's lives.

To resolve remaining uncertainties, further research should prioritize designs that measure individual affectedness directly and explore heterogeneity in responses. A natural progression of my approaches would be to use a measure of flood affectedness based on nature-based indicators, such as precipitation, ground water levels, and soil moisture. In addition, a targeted survey of those who have suffered personal harm would provide insight into whether personal exposure has effects that are not reflected in the regional residence data. Qualitative interviews could uncover why aggregate environmental worry is stagnant and how individuals perceive links between local events and broader climate risks. Additionally, exploiting ASEP (or similar) items that jointly measure climate worry and policy support would allow descriptive and causal analyses of whether and how worry translate into willingness to back specific policies - information essential for effective policy design and communication.

[1] Stefan Obermaißer (Employee, Office of the Lower Austrian Government. Department for Fire Brigades and Civil Defense), in discussion with the author, April 2026

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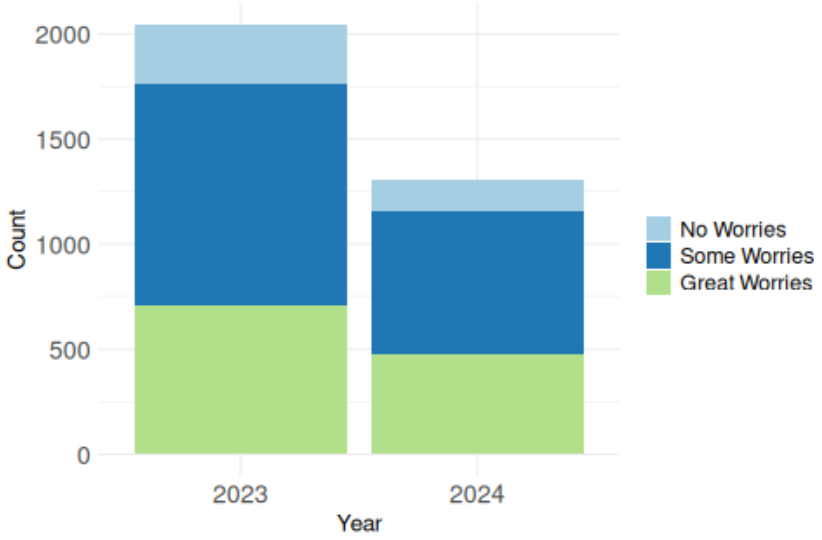
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A. Attrition

Because the ASEP survey is conducted in two waves, there is a risk of attrition bias, which occurs when respondents drop out of the survey between waves in a non-random way. The figure below shows that the sample size decreased by approximately 40% from 2023 to 2024, with a total of 2978 respondents in 2023 and 1845 respondents in 2024.

Figure A.1: Sample 2023 vs. 2024



Source: (ASEP, 2024, 2026)

Table A.1 shows the distribution of climate worry in 2023, comparing those that dropped out (attrition) vs. those that stayed in the survey (retention). Overall, the distribution of climate worry is similar between the two groups, with a slightly higher share of "No Worries" among those that dropped out.

Table A.1: Distribution of Climate Worry in 2023

	Attrition	Retention
No Worries	0.16	0.12
Some Worries	0.52	0.51
Big Worries	0.32	0.36

Source: (ASEP, 2024)

B. Article Search - AI Prompt

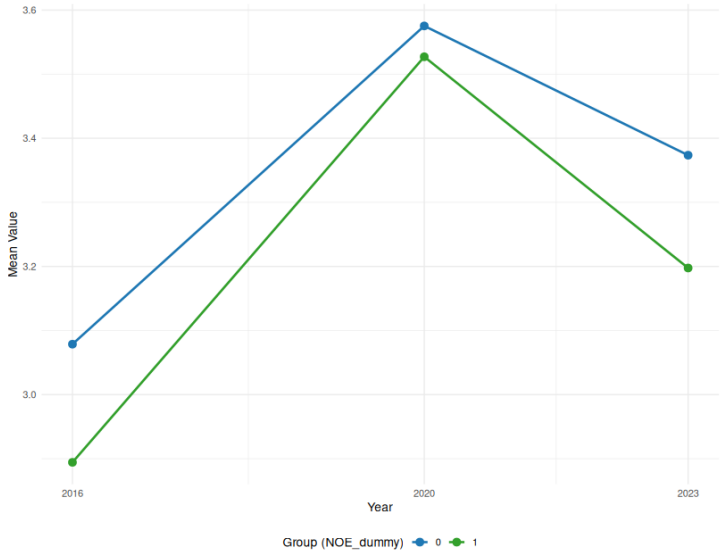
The following prompt was used to extract the relevant municipalities and districts from the news articles about the September 2024 floods in Austria. The AI was tasked with extracting only those locations that were mentioned in the context of an active flood emergency to ensure to include only locations that were actually affected by the flood:

> Analyze this article about the September 2024 floods in Austria.
> EXTRACTION RULES:
>
> 1. TARGET: Extract Austrian Municipalities (Gemeinden) and Districts (Bezirke).
> 2. VIENNA: Map neighborhoods (e.g., Auhof, Hietzing) to their official Vienna District name. For Vienna, use the District as 'town' and 'Vienna' as 'district'.
> 3. ST. PÖLTEN: If the city itself is hit, use town: "St. Pölten (Stadt)". If the surrounding area is hit, use district: "St. Pölten-Land".
> 4. FILTER: Only include locations where the text implies an active flood emergency (e.g., "high water," "evacuation," "flooded streets," "dam risk").
> 5. IGNORE: Do NOT extract towns that are only mentioned as origins of relief forces (e.g., "Firefighters from Villach arrived").

C. Parallel Trends

It is not possible to verify the assumption of parallel trends based on the data underlying the analysis, as the ASEP survey comprises only two waves. I therefore divert to the European Social Survey (ESS) as an alternative. In the ESS, worry about climate change is surveyed as *"How worried are you about climate change?"*, and the answer is measured on a five-point likert scale. For data protection reasons the ESS only includes regional information on the federal state level, which means that I can only compare the trends of Lower Austria to the rest of the country. The following graph shows climate worry for the years 2016, 2020 and 2023, as well as the results from the formal parallel trends test.

Figure C.1: Parallel Trends



	Test Model
(Intercept)	3.196***
	(0.021)
NOE_dummy	-0.124*
	(0.048)
year_num	0.138***
	(0.016)
NOE_dummy × year_num	-0.008
	(0.035)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Both the visual and the statistical test show that the trends of climate worry in Lower Austria and the rest of the country are not significantly different from each other.

D. Logit Summary Output

In addition to the OLS regression output shown in the main text, I also ran an ordered logit regression; the results are shown in table D.1 below.

Table D.1: Ordered Logit Summary Statistics

	Model 1	**Model 2**	**Model 3**	**Model 4**
Post-Treatment	0.062	-0.193	0.022	0.023
	(0.064)	(0.142)	(0.058)	(0.061)
Treated Group	-0.081	-0.308	-0.400	-0.305
	(0.145)	(0.514)	(0.450)	(0.970)
DID (Post x Treated)	-0.226	0.031	-0.166	-0.156
	(0.137)	(0.567)	(0.512)	(1.093)
Num.Obs.	2620	502	2620	2620

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Note: Standard Errors displayed. Controls: age, gender, education, party preference, urban level, anxious character. Cluster-Robust Standard Errors. Estimates displayed as log odds.

Source: (ASEP, 2024, 2026)

E. Sensitivity Checks

Table E.1: OLS results without (1) vs. with controls (2) - based on model (3) from 5.2

	Model 1	Model 2
Post-Treatment	0.007	0.004
Treated Group	-0.330+	-0.136
DID (Post x Treated)	-0.061	-0.065
Gender (0 = male)		0.123***
Age		0.003**
Education		
Compulsory schooling		0.147
Apprenticeship		0.136
Technical College		0.135
Matura		0.239*
Foreman ('Meister')		0.132
University Degree		0.332**
Other education after Matura		0.439***
Party Preference		
SPÖ		0.176***
FPÖ		-0.298***
Greens		0.370***
NEOS		0.203***
Other		0.069
Wouldn't vote		-0.055
Would cast invalid vote		-0.048
Urban Level		
Suburban		-0.061
Rural		-0.071+
Anxious Character		
Sometimes worried (Level 2)		0.045
Often worried (Level 3)		0.159*
Always worried (Level 4)		0.009
Num.Obs.	2620	2620

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Cluster-Robust Standard Errors.

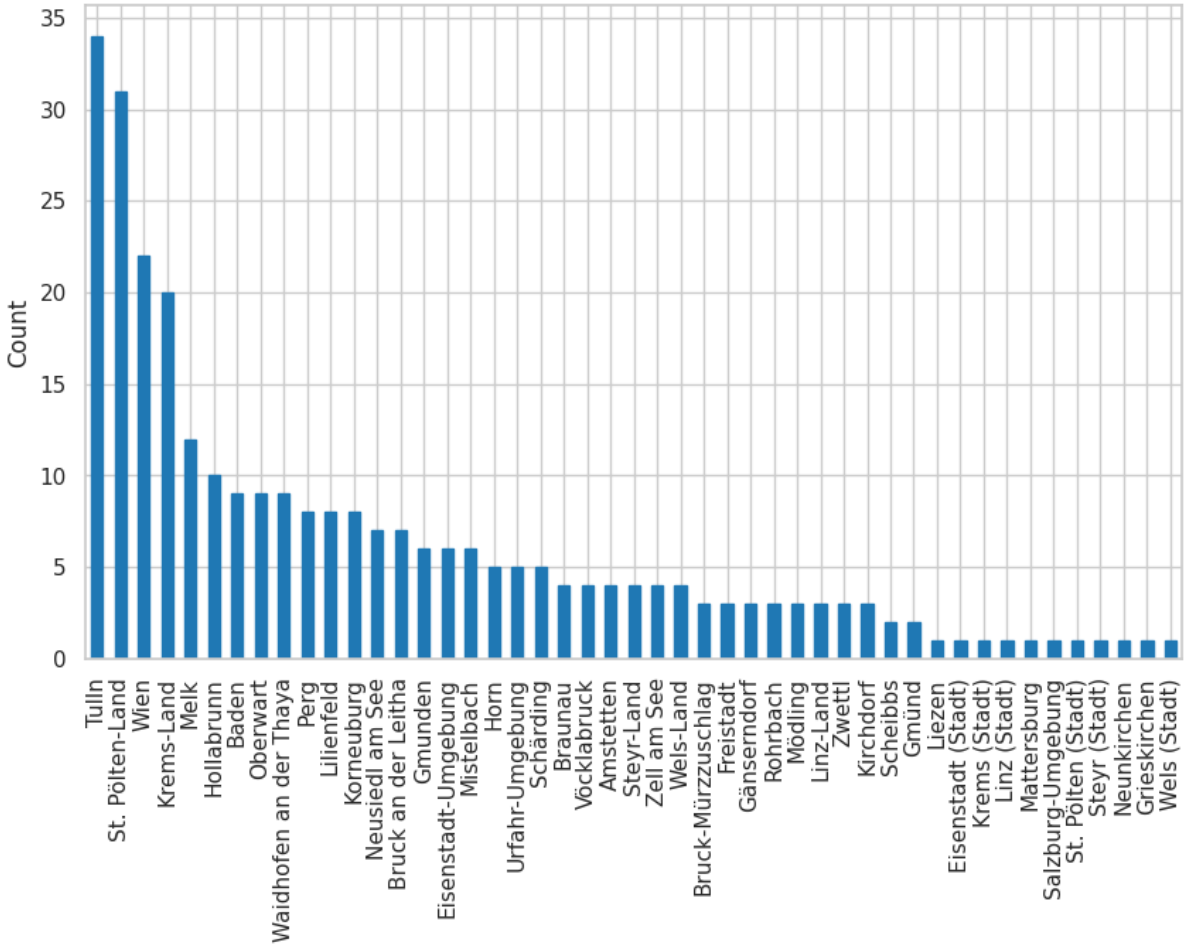
Baseline education: No compulsory schooling. Baseline party preference: ÖVP

Baseline urban level: Urban. Baseline anxious character: Not worried

F. Media Index

Figure F.1 shows the total number of occurrences of the districts in media articles. The municipalities in the Tulln district were mentioned 34 times in media articles, making it the district with the highest media presence. The counts are not weighted.

Figure F.1: Districts - Media Occurences



Source: Article count dataset; assembled by the author

G. Support for Environmental Policies

Table G.1: Willing to accept higher taxes, by party preference

Party Preference	Completely acceptable	Somewhat acceptable	Depends	Rather unacceptable	Completely unacceptable
ÖVP	0.034	0.095	0.247	0.314	0.310
SPÖ	0.031	0.111	0.285	0.308	0.264
FPÖ	0.007	0.034	0.211	0.222	0.525
NEOS	0.184	0.356	0.316	0.090	0.054
Greens	0.005	0.161	0.331	0.255	0.248
Not eligible	0.024	0.099	0.336	0.359	0.182
Wouldn't vote	0.000	0.009	0.239	0.347	0.405
Vote invalid	0.000	0.044	0.243	0.317	0.396

Table G.2: Willing to accept higher prices, by party preference

Party Preference	Completely acceptable	Somewhat acceptable	Depends	Rather unacceptable	Completely unacceptable
ÖVP	0.064	0.195	0.443	0.145	0.153
SPÖ	0.060	0.190	0.354	0.230	0.166
FPÖ	0.003	0.092	0.329	0.235	0.340
NEOS	0.221	0.374	0.317	0.061	0.027
Greens	0.030	0.251	0.413	0.199	0.108
Not eligible	0.046	0.127	0.534	0.190	0.104
Wouldn't vote	0.015	0.042	0.378	0.358	0.207
Vote invalid	0.021	0.031	0.416	0.157	0.375

Table G.3: Willing to accept lower standard of living, by party preference

Party Preference	Completely acceptable	Somewhat acceptable	Depends	Rather unacceptable	Completely unacceptable
ÖVP	0.057	0.214	0.347	0.190	0.192
SPÖ	0.037	0.209	0.397	0.226	0.131
FPÖ	0.000	0.095	0.355	0.215	0.335
NEOS	0.229	0.367	0.340	0.042	0.021
Greens	0.056	0.231	0.446	0.145	0.123
Not eligible	0.019	0.190	0.398	0.273	0.121
Wouldn't vote	0.000	0.105	0.170	0.331	0.394
Vote invalid	0.053	0.184	0.347	0.158	0.258

Individual weights apply

Source: (ASEP, 2026)